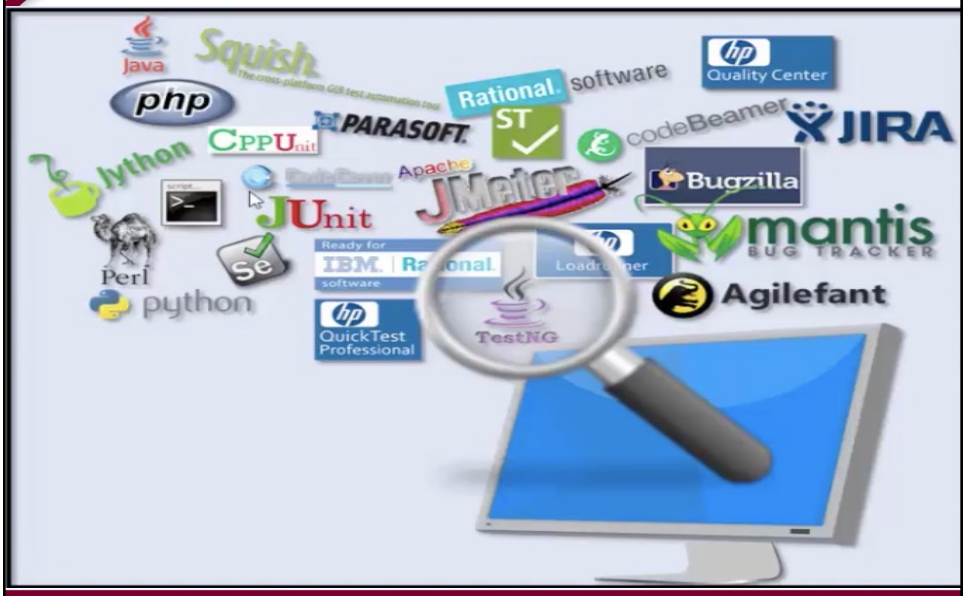
Tools Support for Testing



5

Purpose of Tools Support

- Automate activities
  - Repetitive task like static testing
  - Testing large scale
  - Performance testing
- Support manual test activities
  - Test planning
  - Test Design
  - Test Reporting
  - Test Monitoring



6

Type of Test Tools (1)

- Based on the idea of Computer Assisted Software Engineering (CASE) tools, test tools are also called CAST-Tools (Computer Aided Software Testing).
- Different classifications are possible according to:
  - The supported activity or phase of the test process.
  - The supported test level
  - The supported type of test
  - The supported test role
  - The supported technology
  - Commercial / open-source software/shareware
  - …….

Types of Test Tools (2): Classification according to supported activities

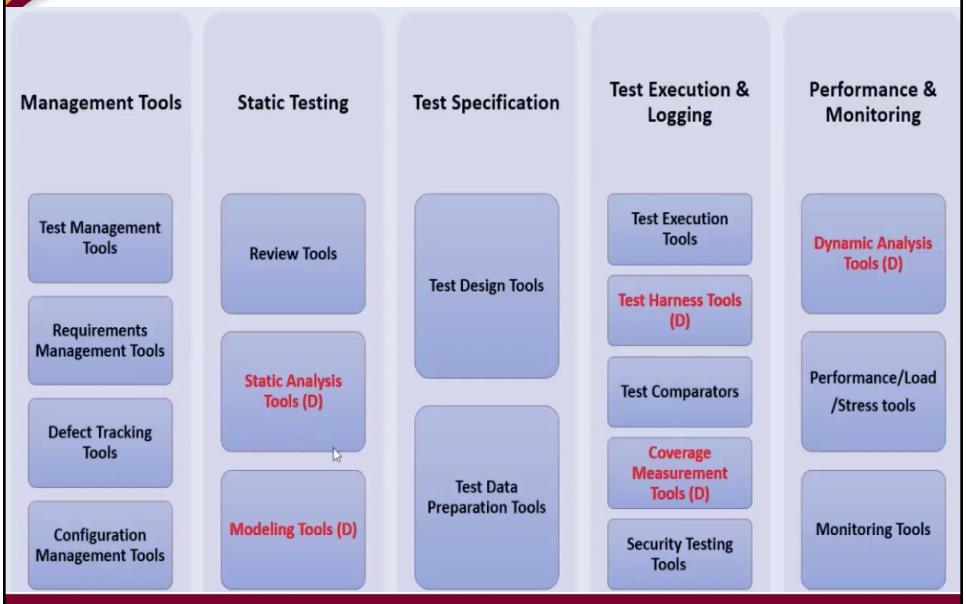
- Test tools according to the supported activity or phase of test process
  - Test planning and control
  - Test analysis and test design
  - Test implementation and execution
  - ….
- Separate test tools support mostly only a small part of the test process.
- Test tool families or test tools suites cover many activities of the test process, or even all activities and can be viewed as an integrated unit.

Types of Test Tools (3): Classification according to main objective group or main user of the tool

- Who should use the tool?
  - Tester?
  - Test manager?
  - Developer, to support the component / integration test?
- Several tools support the software development process
  - Tools especially for the support of the tester.
  - Tools, which are not only used by testers. These tools can also support other activities in the software development process e.g. requirements management.

9

Types of test tools (3)



10

Types of test tools (4)

Management Tools	Static Testing	Test Specification	Test Execution & Logging	Performance & Monitoring
Test Management Tools	Review Tools	Test Design Tools	Test Execution Tools	Dynamic Analysis Tools (D)
Requirements Management Tools	Static Analysis Tools (D)	Test Data Preparation Tools	Test Harness Tools (D)	Performance/Load /Stress tools
Defect Tracking Tools	Modeling Tools (D)		Test Comparators	Monitoring Tools
Configuration Management Tools			Coverage Measurement Tools (D)	
			Security Testing Tools	

11

Test Management Tools

TestRail
kualitee
QAComplete
TestLodge
PractiTest
ZEPHYR
spiraTest
TestLink
qTest Manager
JIRA

12

Test Management: Tools for Test Planning and Control

- Planning tools are used for gathering, cataloging, administration and prioritization test cases.
- Advanced functionality:
  - Import of system requirements
  - Traceability of test, test results and incidents
  - Test case status (e.g: how often executed)
  - Recording of test results and producing of progress reports.
  - Time and resource planning

13

Requirements Management Tools

- Complexity and high number of requirements
  - Need tools for gathering, cataloging, structured storage, change management and prioritizing requirements.
- Advanced functionality:
  - Testing for consistency as well as missing or undefined requirements
  - Traceability of separate test for requirements, functions and/or features
  - Measuring of the degree of requirements coverage, functions, and/or quantity of tests.

14

Incident Management Tools

- Many of defect reports require defect and incident management tools for gathering, administration, sorting and tracking of deviations (e.g. status, change request.)
- Enable the tracking of defects over time.
- Advanced functionality:
  - Parameterizable defect status models (eg. Open, closed)
  - Workflow functionality for the bug fixing process (eg allocation of tasks to determine responsibility for bug fixing)

Incidents Management Tools









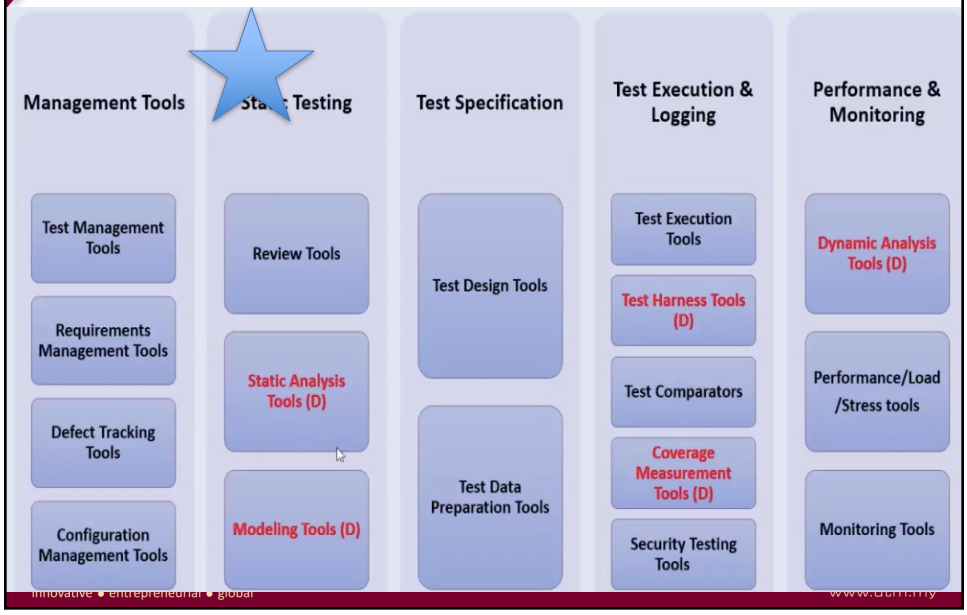








Types of test tools



17

Tools for the Review Process

- Administration of documents (data handling)
  - Administration of reviews and checklists
  - Distribution of review comments and review results
  - Administration and version control of the review documents.
- Supporting the reviewer
- Evaluation (data collection)
  - Supporting the collection and evaluation of review results by means of metrics.
- Supporting the review examination (meeting support)

18

Tools to support for Static Analysis

- Analyse different characteristics of program code.
- Structural characteristics (e.g. cyclomatic number, depth of inheritance etc)
  - Data flow anomalies
  - Adherence of coding conventions.
- Advanced functionality
  - Clone detection
  - Cycle detection (cyclic dependencies between the elements in the source code)

Static Analysis Tools

FindBugs™

raxis information security

CodeScene™ Powered by Empear

PVS-Studio

VISUAL EXPERT

scitools Understand

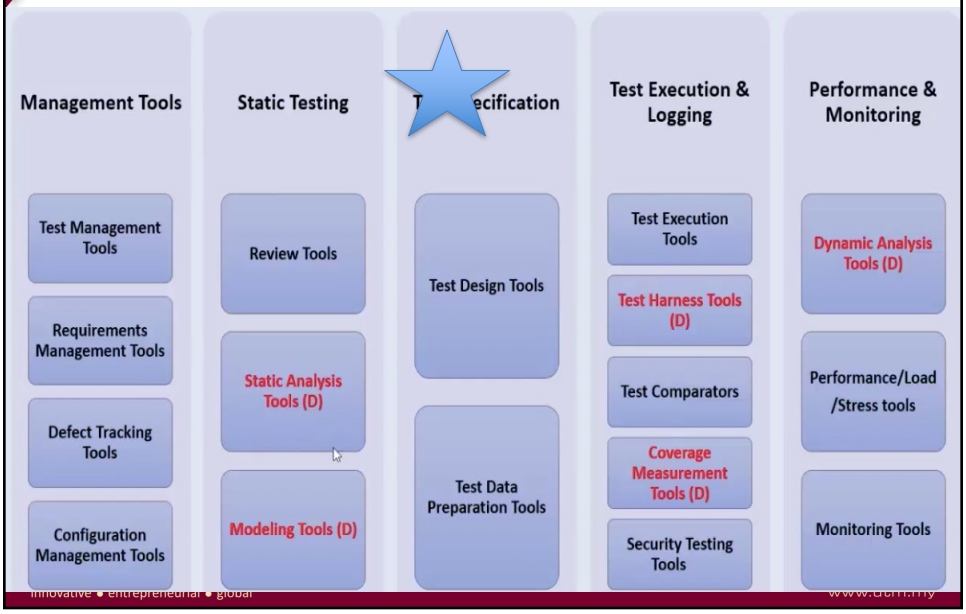
PARASOFT

Modeling Tools

- Support the specifications as well as the validation and verification of models of the software.
- Starting point for the generating test data and test cases from the model.
- Advanced functionality:
  - Code generation
  - Test data and test cases generation
- Main use of static analysis and modeling tools:
  - Effective detection of defects in the development process

21

Type of test tools



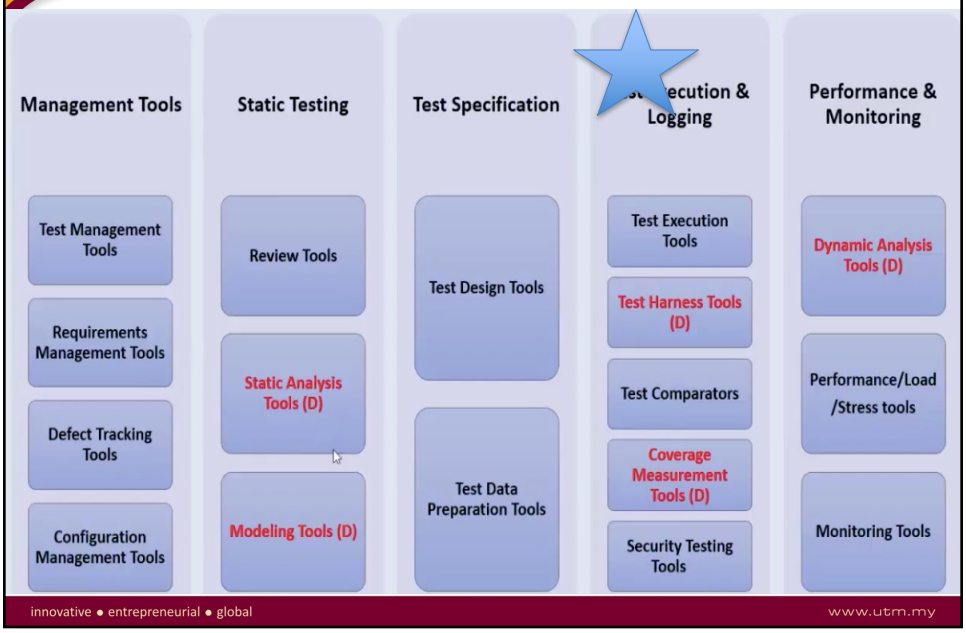
22

Test Design Tools

- Generating test cases or test input data (test data generators) from different models and sources, e.g. from
  - Requirements
  - GUI
  - Design model
  - Code
- Generating test oracle
  - Expected responses
  - Expected post state
- No guarantee for "good" test cases.

23

Type of test tools



24

Test Execution and Recording

- Automated and partially automated versions of test cases:
  - Providing the test object with test data.
  - Recording the response of the test object.
  - Recording the test run.
- The tool must be capable of accessing the test objet, which depends on the test level (component, integration or system test).
- Different types:
  - Test execution tools
  - Simulators
  - Comparators
  - Tools for coverage measurement
  - Security testing tools.

25

Test Execution and Recording (2)

- Automating functional tests.
- Uses the external interface of the test object, for example the user interface.
- Different approaches can be used for the automation of test executions.
  - Capture and replay
  - Script-based testing
  - Data-driven approach
  - Keyword-driven approach

26

Capture & Replay

- Capture of user actions, like mouse movement, mouse clicks and keyboard strokes and saving these actions in scripts.
- Defining and setting of points where expected values will be compared to actual values.
- Replay of the scripts with the check points.
- Test data can often be imported from external sources.
- Testers need to know the specific script language used by the tool, so as to modify the test scripts.
- Capturing can also be a useful support for explorative tests, so that the tester knows the steps executed if a defect is discovered.

Comparators

- Enable the comparison between expected and actual results to be made easily.
- Can handle different file and database formats.
- Test execution tools have comparators for
  - GUI Objects
  - Screen content (eg comparing two screen shots and highlighting the differences)
  - Console content.
- Configuration management tools have comparators for highlighting the difference between two versions of a program.

Automated Testing Tools

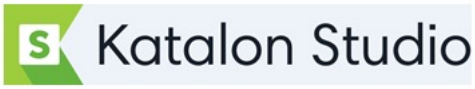














29

Security Testing Tools







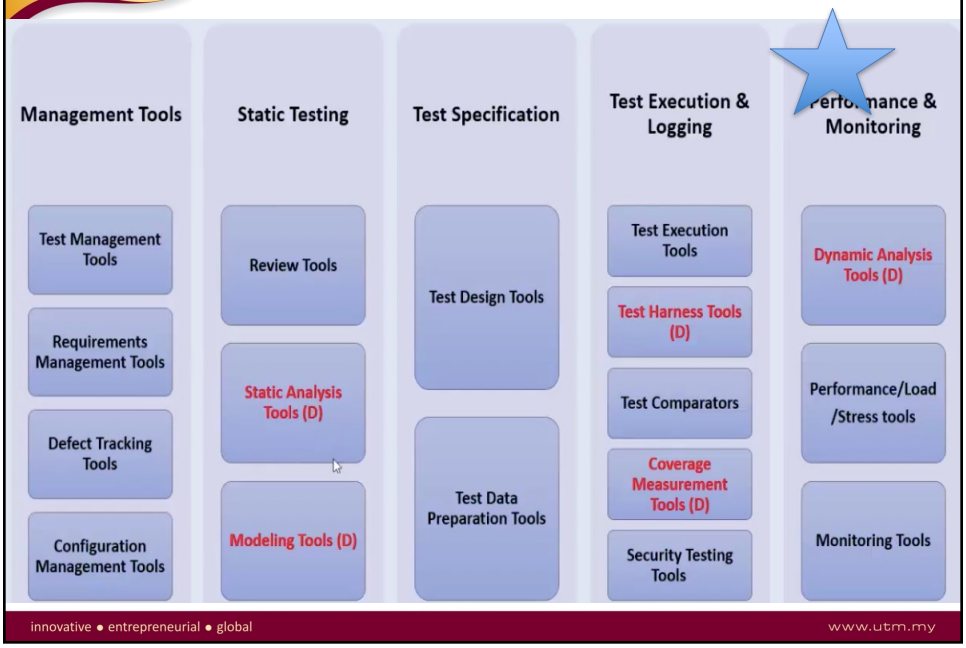






30

Type of test tools



31

Dynamic Analysis Tools

- Provide additional information which can only be gathered when the program runs:
  - Storage use
  - Pointer allocation
  - Pointer arithmetic
  - Memory leaks
- Application: Component and integration test as well as tests for middleware.

32

Performance/Load/Stress Test Tools

- Test of performance-related non-functional requirements.
- Tools for performance tests measure the response time of a system under different simulated usage conditions
  - Number of parallel users.
  - Run-up/ramp-up system behaviour
  - Frequency of transactions.
- A large quantity of virtual users can be simulated to generate the required load.
- A load can be for example:
  - Database queries
  - User transactions, Network traffic

33

Performance Testing Tools



34

Effective Use of Tools: Potential Benefits and Risks

35

Benefits of Using Testing Tools (1)

- Saving of resources by means of efficient processing of tasks.
- Standardisation of the test documentation.
- Simple statistical and graphical evaluation about executed tests eg.:
  - Test process
  - Defect rates
  - Performance
- Unattended

36

Benefits of Using Testing Tools(2)

- A higher degree of automation brings:
  - Consistency and repeatability
  - Traceability of tests
  - More depth of tests
  - Faster execution of large amounts of test possible
  - Relieving the test team from repetitive tasks.
  - Test can be conducted, which would be very difficult or impossible if executed manually.
  - High reliability of test, for example, by means of automated comparison of high data volume.

Risks of Testing Tools (1)

- Unrealistic expectations towards automation and tool.
- Underestimation of time, costs and effort.
  - From initial implementation of the tool (including training and consultancy).
  - To adjust existing processes.
  - For the maintenance of the artefacts, which have been created by the tool.
- Blind trust into the tool
  - Neglect or even replacement of test design.
  - Replacement of tests that were actually much more adequate as manual tests.

Risks of Testing Tools (2)

- The tool vendor changes
- Risks of insufficient support from the vendor
- Neglect of the complexity of interfaces to other tools.

39

Choice and Introduction of Test Tools into an Organization

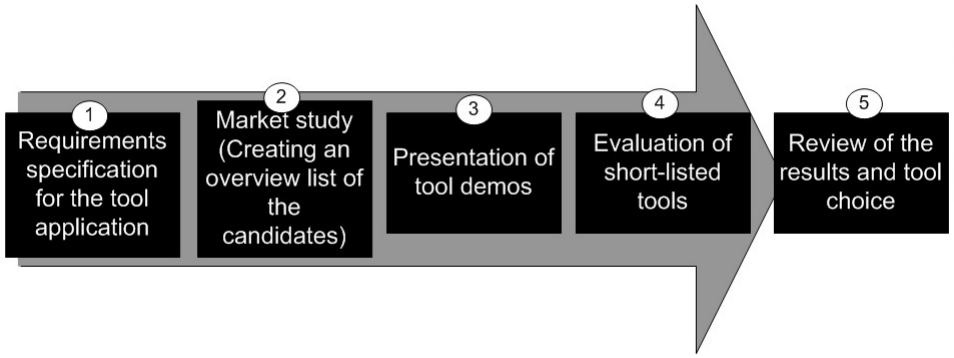
40

Costs of Tool Implementation

- When implementing a tool, costs (mostly) arise from:
  - Selection
  - Purchase
  - Maintenance
  - Hardware
  - Staff training

41

Process of Tool Selection



42

Criteria for **Requirements Specification**

- How well the tool communicates with the potential test object
- Know-how of the tester about tools and methods
- Possibility for integration into the existing development environment.
- Possibility for integration with other testing tools to be used.
- Platform on which the tool should run.
- Service, training offered, reliability and market position of the vendor.
- License conditions, price, maintenance costs.

Market Study and **Tool Demos**

- Creating a list of available tools in a specific category.
- Limitation to a short-listed of candidates based on the criteria catalogue and requested information about the tools or internet research.
- Tool demonstration of the vendor of short-listed products.

Evaluation and Tool Selection

- After tool demo, the following main points are to be verified in the final evaluation round:
  - Does the tool work with the test objects and the development environment?
  - Were other performance characteristics can be achieved in practice?
  - Can the vendor deliver support for non-standard questions?

45

Selecting a Tool

- Select the right tool to succeed at test automation
- Perform a realistic assessment
- Define clear requirements and objective criteria
- Do at least one proof-of-concept
- Evaluate vendor
- Evaluate training
- Plan for coaching and mentoring
- Understand the business case

Copyright © RBCS 1996-2911

46

Objectives of an Automation Pilot Project

- Learn more about the tool and how to use it
- Adapt the tool and/or processes and practices to fit organisation and systems
- Standardise ways of using, managing, storing and maintaining the tool and the test assets
- Assess the ROI (return of investment)
  - For e.g. cost benefit analysis compared with manual testing

Copyright © RBCS 1996-2911

Success Factors for Tool Deployment

- Deploy the tool to the rest of the organisation incrementally (repeat cost benefit analysis at every cycle, and produce lessons learned document)
- Adapt and improve the test processes to fit use of the tool
- Provide training, coaching and mentoring for new users
- Define tool usage guidelines
- Learn ways to improve use of the tool continuously
- Provide support, coaching and mentoring
- Gather lessons learned
- Monitor tool use and benefits

Copyright © RBCS 1996-2011